

ORIGINAL ARTICLE

Lifetime Adverse Pregnancy Outcome History and Cardiovascular Risk

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BACKGROUND: Few studies have examined how multiple types of adverse pregnancy outcomes across women's reproductive lives relate to long-term cardiovascular disease.

METHODS: In 59 154 parous participants in Nurses' Health Study II, lifetime history of gestational diabetes, gestational hypertension, preeclampsia, preterm delivery, and low birthweight was self-reported, and cardiovascular events, including myocardial infarction, stroke, and coronary revascularization, were identified through June 2017. We used Cox proportional hazards models to estimate associations between adverse pregnancy outcomes and cardiovascular disease and quantified the extent to which these associations were explained by the later development of hypertension, diabetes, and hypercholesterolemia. We evaluated whether adding adverse pregnancy outcomes improved prediction of premature cardiovascular disease beyond established risk factors such as systolic blood pressure and diabetes.

RESULTS: Each adverse pregnancy outcome was associated with a higher risk of long-term cardiovascular disease. Only gestational hypertension (hazard ratio, 1.62 [95% CI, 1.36–1.92]) and preeclampsia (1.31 [1.11–1.55]) retained independent associations after accounting for the cooccurrence of other adverse pregnancy outcomes. Postpregnancy hypertension, diabetes, and hypercholesterolemia jointly accounted for substantial attenuation (58.4% [38.7%–75.8%]) of the association between adverse pregnancy outcomes in the first pregnancy and later cardiovascular disease. Adding adverse pregnancy outcomes only modestly improved discrimination and slightly improved reclassification.

CONCLUSIONS: Common adverse pregnancy outcomes, especially gestational hypertension and preeclampsia, are associated with higher future cardiovascular risk, with much of this association attenuated after accounting for subsequent cardiovascular risk factors. However, given the limited predictive gains, more nuanced integration of adverse pregnancy outcomes is needed to enhance their clinical utility in cardiovascular risk prediction. (*Hypertension*. 2026;83:00–00. DOI: 10.1161/HYPERTENSIONAHA.125.25916.) • **Supplement Material.**

Key Words: blood pressure ■ humans ■ pregnancy ■ risk factors ■ stroke

Adverse pregnancy outcomes (APOs), such as hypertensive disorders of pregnancy (HDP; preeclampsia and gestational hypertension [GHTN]), gestational diabetes (GDM), low birth weight (LBW), and preterm delivery (PTD) affect around one-third of pregnancies^{1,2} and are on the rise.³ Substantial evidence consistently links APOs to elevated risks of not only future cardiovascular disease (CVD) but also established CVD risk factors, such as hypertension, diabetes, and hypercholesterolemia in women.⁴ However, while multiple APOs

can occur over a woman's reproductive life stage, their independent and joint associations with future CVD risk, as well as the potential of lifetime APO history to improve CVD risk prediction, are poorly understood.

Few studies have examined multiple APOs within single populations in relation to CVD events, and their findings are inconsistent.^{1,2,5–9} For example, the WHI (Women's Health Initiative) in the United States found HDP and LBW to be independently associated with CVD, while GDM and PTD were not, and did not assess interactions

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NOVELTY AND RELEVANCE

What Is New?

Our study is among the few to integrate multiple types of adverse pregnancy outcomes (APOs) across the reproductive lifespan in one of the largest, longest-running, and well-characterized cohorts of women. We examined the long-term cardiovascular consequences of APOs over >35 years of follow-up, quantified the statistical mediating role of cardiovascular risk factors, and evaluated the additive predictive value of APOs for cardiovascular disease beyond traditional risk factors.

What Is Relevant?

Common APOs, including gestational hypertension, preeclampsia, gestational diabetes, low birth weight, and preterm delivery, are early signals of heightened cardiovascular risk decades later. Much of this excess risk is attenuated upon adjustment of subsequent development of hypertension and diabetes. However,

adding APOs to existing cardiovascular risk calculators provided only small improvements in predictive performance, suggesting the need to explore more effective strategies for developing APO-based cardiovascular disease risk prediction tools.

Clinical/Pathophysiological Implications?

APOs are important women-specific signal of future cardiovascular disease that often precede the onset of cardiovascular disease risk factors including chronic hypertension or diabetes. Clinicians should consider incorporating reproductive history into cardiovascular risk assessments and provide proactive follow-up and preventive care for women with a history of APOs, even in the absence of overt cardiometabolic disease at postpartum visits. Tools that estimate short- and long-term cardiovascular risk after pregnancy are needed to better guide timely and targeted prevention strategies during the postpartum period and beyond.



Nonstandard Abbreviations and Acronyms

APO	adverse pregnancy outcome
CVD	cardiovascular disease
GDM	gestational diabetes
GHTN	gestational hypertension
HDL	high-density lipoprotein
HDP	hypertensive disorders of pregnancy
HR	hazard ratio
LBW	low birth weight
MI	myocardial infarction
NHS II	Nurses' Health Study II
PTD	preterm delivery
WHI	Women's Health Initiative

between APOs.⁵ In contrast, a national Swedish cohort study found that GDM, HDP, and PTD were independently associated with ischemic heart disease² and CVD mortality,⁸ with evidence of additive interactions between APOs. Notably, only one of these studies focused on a United States population,⁵ and none of them examined the potential role of established CVD risk factors in the APOs-CVD association.

While adding individual APOs to models of established CVD risk factors has shown limited improvement in CVD risk prediction,^{10–14} it is unclear whether including multiple APOs would enhance predictive performance. Of the 2 studies addressing this question, 1 examined eclampsia and GDM in a United Kingdom population

with much lower APO prevalence (<0.1% for eclampsia, 0.3% for GDM) than is typical,¹⁵ and the other examined HDP and GDM in a United States population with a follow-up period of just 5 years.⁷ Both studies did not begin CVD risk prediction until ≈10 years after women's first pregnancy when they were over age 40 or 45, resulting in potentially missed opportunities for early prevention.

We leveraged data from the Nurses' Health Study II (NHS II) to address these research gaps through 3 objectives. First, we estimated the individual, independent, and joint associations between lifetime APO history and long-term risks of CVD events. Second, we assessed the extent to which the association between APO and CVD events was affected by established CVD risk factors arising for the first time after pregnancy. Third, we determined whether adding multiple types of APOs improved the prediction of premature CVD risk beyond established CVD risk factors in women ≥30 or 40 years.

METHODS

The data that support the findings of this study are available from the NHS II committee upon reasonable request.

Study Population

NHS II is an ongoing prospective cohort of 116 429 female registered nurses from 14 US states who were aged 25 to 42 years at enrollment in 1989.¹⁶ Questionnaires were administered at baseline and biennially thereafter to collect information on diet, lifestyle, medical history, and health status. This study was approved by the Mass General Brigham Human Research

Committee of Brigham and Women's Hospital. Questionnaire return implied informed consent.

APO Assessment

In the 2009 NHS II questionnaire, 76 839 participants reported their complete lifetime pregnancy history, including gestational length, birth weight, and complications for each pregnancy. Compared with nonrespondents ($n=39\,590$), participants who responded to the 2009 questionnaire were largely similar at baseline in 1989, except that they were more likely to be White, have higher parental education and a parental history of CVD, and were less likely to be current smokers (Table S1).

We focused on 5 APOs in the current study: GHTN, preeclampsia, GDM, LBW, and PTD. Birth weight, which was reported in categories (<5 , 5–5.4, 5.5–6.9, 7–8.4, 8.5–9.9, ≥ 10 lbs), was used to define LBW as <2500 g (<5.5 lbs). Gestational length was also reported in categories (<8 , 8–11, 12–19, 20–27, 28–31, 32–36, 37–39, 40–42, ≥ 43 weeks). We classified PTD as very preterm (20 to <32 weeks), moderate to late preterm (32 to <37 weeks), or no PTD (≥ 37 weeks). GHTN was defined based on self-reported pregnancy-related high blood pressure, and preeclampsia was defined as self-reported preeclampsia/toxemia. In previous validation studies conducted in subgroups of NHS II participants, the positive predictive value of self-reported preeclampsia compared with medical records ranged from 69% to 89%,^{10,17} and that for self-reported GDM was 94%;¹⁸ the kappa statistic for self-reported PTD characterized in 3 categories was 0.74.¹⁹ We only included pregnancies that lasted ≥ 20 weeks, and excluded participants who did not complete the 2009 questionnaire, were nulliparous in 2009, were <18 or >45 years at first pregnancy, or lacked the year or age of first pregnancy (Figure S1). We were unable to include small for gestational age as one of the APOs of interest because both birth weight and gestational length were reported in categories rather than continuous values.

CVD Ascertainment

Myocardial infarction (MI), stroke, and coronary revascularization reported through June 2017 (ie, the end of this study) comprised the CVD outcomes of interest. On biennial questionnaires, participants reported any physician-diagnosed MI or stroke since 1989 and coronary revascularization since 1995. MI or stroke (both fatal and nonfatal) were confirmed by medical record reviews or corroborated by supplemental questionnaires from participants or next of kin, as detailed elsewhere.^{10,19} Because revascularization was not separately confirmed, we conducted a validation study among 71 NHS II participants who reported both MI and coronary revascularization in the same questionnaire cycle. Medical records confirmed revascularization in 66 participants, and the remaining 5 all had documentation of transfer for catheterization without available records from the receiving hospital. We excluded participants with CVD events before their first pregnancy, or who reported MI or stroke at baseline in 1989 because these events cannot be confirmed or precisely dated (Figure S1).

Measurement of Other Variables

Race/ethnicity was self-reported in 1989. Parental education was reported in 2005, and history of MI or stroke before age

60 was reported in 1989, 1997, 2001, and 2013. Biennial questionnaires provided information on body mass index, smoking, and diagnosis and treatment of hypercholesterolemia, diabetes, and hypertension. Prior validation studies have demonstrated the accuracy of participants' self-reported hypercholesterolemia²⁰ (positive predictive value: 86% in a similar nurse cohort), diabetes²¹ (positive predictive value: 84%), and hypertension²² (sensitivity: 94%, specificity: 85%). Body mass index for each age from 18 to 67 was derived via a published method.¹⁷ Alcohol intake was assessed on every other questionnaire cycle. Physical activity before first pregnancy was based on reported frequency of strenuous activity at ages 18 to 22 in 1989. Age and parity for each pregnancy were captured by the 2001 and 2009 questionnaires. Systolic blood pressure was reported in categories (<105 , 105–114, 115–124, 125–134, 135–144, 145–154, 155–164, 165–174, ≥ 175 mm Hg) in 1989, 1999, 2005, 2009, and 2013; we assigned the midpoint of each category as the continuous systolic blood pressure value. Plasma total and HDL (high-density lipoprotein) cholesterol were predicted at each biennial cycle using published methods, which correlated well (correlation coefficients: 0.50–0.57) with measured values available in a subset of NHS II participants.^{10,13}

Statistical Analyses

For association estimation, we used Cox proportional hazards models with first delivery as the time origin and age as the time scale. Participants were followed until a first CVD event, last questionnaire return, death (confirmed by the National Death Index), or the end of the study (2017 questionnaire cycle), whichever occurred first. The oldest participants reached age 73 years by the end of follow-up, which served as the upper bound for cumulative incidence and hazard ratio (HR) estimation. We accounted for within-individual correlations arising from multiple pregnancies using robust sandwich variance estimates. We additionally excluded a small portion of participants with chronic hypertension or diabetes before their first pregnancy to improve the validity of HDP and GDM classification (Figure S1), as under earlier diagnostic criteria women with these preexisting chronic conditions were not eligible to be diagnosed with HDP or GDM. We adjusted for potential confounders before the first pregnancy, including parental education, parental history of MI or stroke before age 60 years, body mass index, smoking, alcohol intake, high cholesterol, and physical activity; additionally, we adjusted for parity, which was updated at the time of each pregnancy. We applied a missing indicator approach to handle incomplete confounder data (percentages are reported in Table 1).²³

We modeled APOs as time-varying exposures in 2 ways. First, APOs were treated as ever versus never, with participants transitioning from unexposed to exposed at the time of their first APO. Each type of APO was assessed separately and then together to assess their individual and independent associations with CVD. We tested all possible 2-way interactions between APOs on both additive and multiplicative scales. For additive interaction, we calculated the relative excess risk due to interaction with 95% bootstrapped CIs. For multiplicative interaction, we used likelihood ratio tests comparing models with and without interaction terms. Second, we modeled APOs as cumulative episodes (0, 1, 2, or ≥ 3), regardless of the type



Table 1. Age-Standardized Characteristics of Nurses' Health Study II Participants Before Their First Pregnancy

Characteristics	Ever APO, n=17 368	Never APO, n=41 786	Overall, n=59 154
Age at first pregnancy, y; mean (SD)*	26.8 (4.8)	26.6 (4.6)	26.6 (4.7)
White, n (%)	16 212 (93.3)	39 451 (94.4)	55 660 (94.1)
Participants' mother's education, n (%)			
<9 y	1105 (7.3)	2451 (6.6)	3558 (6.8)
9–11 y	1751 (11.6)	4180 (11.2)	5933 (11.3)
12 y	7547 (49.8)	18 547 (49.8)	26 091 (49.8)
13–15 y	3372 (22.3)	8510 (22.9)	11 882 (22.7)
≥16 y	1375 (9.1)	3551 (9.5)	4925 (9.4)
Participants' father's education, n (%)			
<9 y	1733 (11.6)	3822 (10.4%)	5558 (10.8)
9–11 y	1917 (12.8)	4795 (13.1%)	6711 (13.0)
12 y	5716 (38.3)	14 230 (38.8%)	19 946 (38.6)
13–15 y	2602 (17.4)	6093 (16.6%)	8700 (16.8)
≥16 y	2973 (19.9)	7770 (21.2%)	10 737 (20.8)
Parental history of MI or stroke, n (%)	4690 (27.0)	9714 (23.2)	14 407 (24.4)
Strenuous activity at ages 18–22, n (%)			
Never	4736 (27.4)	11 781 (28.3)	16 525 (28.1)
1–3 mo/y	5453 (31.6)	13 136 (31.6)	18 592 (31.6)
4–6 mo/y	3063 (17.8)	7069 (17.0)	10 124 (17.2)
7–9 mo/y	2030 (11.8)	4859 (11.7)	6888 (11.7)
10–12 mo/y	1973 (11.4)	4713 (11.3)	6684 (11.4)
Prepregnancy BMI, kg/m ² ; mean (SD)	22.4 (3.4)	21.8 (2.7)	22.0 (2.9)
Prepregnancy smoking status, n (%)			
Never	11 511 (68.1)	28 175 (68.9)	39 693 (68.7)
Former	1478 (8.7)	3794 (9.3)	5270 (9.1)
Current	3917 (23.2)	8915 (21.8)	12 826 (22.2)
Prepregnancy alcohol intake, g/d, mean (SD)	3.5 (6.3)	3.5 (5.9)	3.5 (6.0)
Prepregnancy high cholesterol, n (%)	552 (3.2)	927 (2.2)	1490 (2.5)

Missingness in the overall sample: maternal education 11.4%, paternal education 12.7%, strenuous activity at ages 18–22 0.6%, prepregnancy BMI 2.6%, prepregnancy smoking 2.3%. APO indicates adverse pregnancy outcome;

BMI, body mass index; and MI, myocardial infarction.

*Value is not age standardized to the age distribution of the study population.

of APO or whether they occurred during the same or different pregnancies, to capture the cumulative risks associated with increasing APO episodes.

Because pregnancies in our sample occurred over a wide time span (1964–2010), we assessed potential cohort effects by examining whether associations between APOs and CVD differed by calendar period. We stratified analyses by pregnancies occurring before versus in or after 1985 (when ≈50% of pregnancies occurred before this year). We obtained *P*-interaction by comparing models with and without the interaction terms between each APO and pregnancy era.

To estimate the impact of established CVD risk factors arising after pregnancy, we estimated the proportion of the association between APOs in the first pregnancy and CVD events that was attenuated by hypertension, diabetes, and hypercholesterolemia occurring after the first pregnancy. We conducted analyses for each individual APO type, any APO, and cumulative APO episodes. Women with hypercholesterolemia, hypertension, or diabetes before their first pregnancy were excluded from this analysis (Figure S1). Each intermediate was considered present from the time of diagnosis through the end of follow-up. We fitted Cox proportional hazards models with and without the intermediates, and estimated the proportion mediated with 95% CIs using the %mediate SAS macro developed by internal NHS II investigators.²⁴

For risk prediction performance comparison, we used Cox proportional hazards models to predict CVD risk in women aged ≥30 years or ≥40 years, separately. We chose to begin follow-up at ages 30 and 40 years in this particular analysis for 3 reasons. First, given their known associations with CVD risk factors already included in prediction models, we hypothesize that APOs are more likely to improve risk prediction in the years before CVD risk factors have arisen. Prior studies on related topics have primarily focused on older populations typically aged 45 years and above. Second, considering young adult predictors of later chronic disease can yield a longer runway for prevention if high-risk individuals can be identified early. With this in mind, age 30 years captures the early reproductive years, while age 40 years marks the end of the reproductive period for most participants. Third, this approach aligns with our research group's previous work on PTD and CVD risk prediction.¹³ We defined established CVD risk factors as variables included in both the Pooled Cohort Equation-Atherosclerotic Cardiovascular Disease calculator²⁵ and the American Heart Association Predicting Risk of CVD EVENTS equation,²⁶ namely systolic blood pressure, treatment of hypertension, total cholesterol, HDL, smoking, and diabetes history.

We compared a Base Model of these established factors with 3 expanded models: Expanded Model 1 added parity and APOs found to be independently associated with CVD in Aim 1, Expanded Model 2 added parity and the cumulative episodes of these APOs, and Expanded Model 3 added the cumulative episodes of any APO. Information on risk factors and APOs was assessed up until age 30 or 40 years. We evaluated calibration based on the slope of the regression line comparing deciles of predicted and observed risks. We assessed discrimination using the Harrell C statistic²⁷ and its changes. We evaluated reclassification using the net reclassification index at the event rate²⁸ to avoid arbitrary selection of risk category thresholds and to facilitate comparisons across populations or study designs. CIs were calculated via 1000 bootstrap samples. Because a cutoff age is required to define the time horizon for observed risk when evaluating calibration and net reclassification index, we selected age 65 years instead of age 73 years (the oldest age at the end of follow-up). Age 65 years is commonly used to define premature CVD in women, and this threshold captured the majority of events (≈89%) while retaining a sufficient proportion of participants under observation (≈40%) for reliable estimation of observed risk. The event rate by age 65 years was 3.3% in the sample used for prediction starting at age

30 years, and 3.1% in the sample used for prediction starting at age 40 years.

Sensitivity analyses included (1) repeating association analyses using only the first pregnancy, rather than the entire pregnancy history, (2) excluding person-time between the time origin and study entry (1989) to address immortal person-time for all analyses, and (3) excluding coronary revascularization from the composite outcome, as it was less rigorously validated than MI or stroke. We assessed the proportional hazards assumption for all Cox models using the Schoenfeld residual-based tests.²⁹ We determined statistical significance at a 2-sided $P < 0.05$ level and conducted all analyses in R 4.2.0 unless otherwise specified. The conceptual framework of the analytic approach is shown in Figure S2.

RESULTS

Participant Characteristics

A total of 59 154 participants contributed to this analysis. Characteristics before the first pregnancy were generally similar between women who ever (29.4%) versus never (70.6%) had an APO (Table 1). Most participants were White, had parents with at least a high school level education, and 24% had a parental history of MI or stroke before age 60 years, with a somewhat higher proportion (27%) among those with an APO history. The majority engaged in some strenuous physical activity at ages 18 to 22 years; more than half had never smoked, and only 2.5% reported high cholesterol before their first pregnancy. Mean prepregnancy body mass index was 22.0 (SD, 2.9) kg/m², and the mean alcohol intake was 3.5 (SD, 6.0) g/d.

Among participants, 7.0% had a history of GHTN, 8.0% preeclampsia, 5.6% GDM, 8.8% LBW, and 13.8% PTD. Overall, 15.4% of participants experienced 1 APO episode, 8.5% had 2, and 5.5% had ≥ 3 , irrespective of the APO type or whether they occurred during the same or different pregnancies (Figure S3). Among those with 2 APO episodes, 57.8% had both APOs in the same pregnancy and 42.2% in different pregnancies; of the latter, 55.9% involved APOs of the same type and 44.1% involved different types. For participants with ≥ 3 APO episodes, 34.7% had all APOs in the same pregnancy, while 65.3% experienced them across multiple pregnancies, with 17.0% involving the same type of APO and 83.0% involving different types.

We observed 1524 incident CVD events during 2105, 441 person-years over a mean follow-up of 35.6 (SD, 7.4; range, 1–53) years. These included 25.3% MI, 31.1% stroke, and 43.6% coronary revascularization events. The mean age at event was 55.8 (SD, 7.5) years, with a median of 56.6 (Q1, 50.8; Q3, 61.5) years. The mean calendar year at event was 2008 (SD, 6), with a median of 2010 (Q1, 2004; Q3, 2014). These distributions were similar across event types (Figure S4).

APO-CVD Associations

Individually, each APO was associated with a higher CVD hazard: GHTN (HR, 1.84 [95% CI, 1.57–2.15]), preeclampsia (1.60 [1.37–1.87]), GDM (1.26 [1.03–1.55]), LBW (1.31 [1.12–1.54]), moderate to late PTD (1.18 [1.00–1.38]), and very PTD (1.38 [1.09–1.75]; Table 2). However, in the model including all APOs simultaneously, HRs for each APO were reduced, and only GHTN (1.62 [1.36–1.92]) and preeclampsia (1.31 [1.11–1.55]) remained independently associated with CVD. We observed no statistically significant additive or multiplicative interactions between APOs (Table S2). Having more cumulative APO episodes was associated with increasingly higher CVD hazards: 1.28 (1.11–1.46), 1.47 (1.25–1.73), and 1.92 (1.59–2.30) for 1, 2, and ≥ 3 episodes, respectively.

To contextualize these associations in terms of absolute risk, we plotted cumulative incidence of CVD over age (up to 73 years, the maximum age at the end of follow-up) by each APO type and by cumulative episodes of any APO (Figure). Both observed and adjusted cumulative incidence curves showed higher absolute CVD risk among women with any APO exposure, with the greatest separation observed for GHTN, preeclampsia, and higher cumulative APO burden. Absolute risk differences were attenuated after adjustment for confounders and APO cooccurrence.

In stratified analyses by pregnancy era, GHTN, preeclampsia, and PTD had slightly higher HRs in pregnancies before 1985, whereas GDM had slightly higher HRs in pregnancies in or after 1985, and LBW had similar HRs regardless of era (Table S3). A statistically significant cohort effect was observed only for GHTN (Figure S5). Specifically, in pregnancies occurring before versus on or after 1985, the HRs were 2.15 (1.76–2.63) versus 1.58 (1.23–2.02) after adjustment for confounders, and 1.88 (1.50–2.35) versus 1.40 (1.06–1.83) after further adjustment for APO cooccurrence (both P -interaction < 0.05). Consistent with these findings, cumulative incidence of CVD was also higher among women with GHTN-complicated pregnancies before 1985.

Statistical Mediation by Hypertension, Diabetes, and Hypercholesterolemia

Comparing women who developed any versus none of the 5 APOs in the first pregnancy, adjustment for subsequent hypertension, diabetes, and hypercholesterolemia was associated with a 58.4% (38.7%–75.8%) attenuation of the APO-CVD association (Table 3). The degree of attenuation increased with the number of APO episodes experienced in the first pregnancy. For individual APO types, adjustment for these factors was associated with a 71.9% (41.6%–90.2%) attenuation of the association for GHTN, 60.8% (38.3%–79.4%)

Table 2. Associations Between Adverse Pregnancy Outcomes and Incident Cardiovascular Diseases

Adverse pregnancy outcomes	n (%)	Cases/person-years	Incidence rate (per 10000 Person-years)	Hazard ratio (95% CI)	
				Association of Individual APO with CVD*	Association of APO with CVD after adjusting for other APOs†
Gestational hypertension‡					
Never	55 008 (93.0%)	1341/1 971 098	6.8	Reference	Reference
Ever	4146 (7.0%)	183/1 34 343	13.6	1.84 (1.57–2.15)	1.62 (1.36–1.92)
Preeclampsia					
Never	54 444 (92.0%)	1335/1 945 221	6.9	Reference	Reference
Ever	4710 (8.0%)	189/160 220	11.8	1.60 (1.37–1.87)	1.31 (1.11–1.55)
Gestational diabetes					
Never	55 847 (94.4%)	1425/2 006 343	7.1	Reference	Reference
Ever	3307 (5.6%)	99/99 098	10.0	1.26 (1.03–1.55)	1.16 (0.94–1.43)
Low birth weight					
Never	53 609 (90.6%)	1331/1 916 525	6.9	Reference	Reference
Ever	5203 (8.8%)	179/175 820	10.2	1.31 (1.12–1.54)	1.18 (0.98–1.42)
Preterm delivery					
Term	50 899 (86.0%)	1273/1 829 547	7.0	Reference	Reference
Preterm: 32-37 wk	6111 (10.3%)	173/202 759	8.5	1.18 (1.00–1.38)	1.05 (0.88–1.26)
Preterm: <32 wk	2094 (3.5%)	75/71 523	10.5	1.38 (1.09–1.75)	1.22 (0.93–1.60)
Cumulative APO episodes§					
None	41 786 (70.6%)	950/1 523 889	6.2	Reference	
1 episode	9086 (15.4%)	267/313 669	8.5	1.28 (1.11–1.46)	
2 episodes	5014 (8.5%)	171/167 752	10.2	1.47 (1.25–1.73)	
≥3 episodes	3268 (5.5%)	136/100 131	13.6	1.92 (1.59–2.30)	

All models adjusted for participants' mother's and father's education (<9, 9–11, 12, 13–15, ≥16 y [reference]), parental history of myocardial infarction or stroke before age 60 years (no [reference], yes), prepregnancy body mass index (continuous), prepregnancy smoking (never [reference], former, current), prepregnancy alcohol intake (continuous), prepregnancy high cholesterol (no [reference], yes), strenuous activity at ages 18–22 years (never, 1–3 [reference], 4–6, 7–9, 10–12 mo/y), and parity (continuous). Missingness: low birth weight 0.6%, preterm delivery 0.2%. APO indicates adverse pregnancy outcome; and CVD, cardiovascular disease.

*Separate models included one type of APO at a time.

†Model included all types of APOs simultaneously.

‡Associations differed between pregnancies occurred before vs on or after 1985. See Figure 4 for estimates.

§Regardless of the type of APO or whether they occurred during the same or different pregnancies.

for preeclampsia, and 29.9% (10.9%–59.9%) for LBW. For PTD, the attenuation was 51.1% (0.1%–100.0%) for births at 32 to 37 weeks and 21.8% (12.4%–35.5%) for births before 32 weeks. Among the 3 established CVD risk factors, adjustment for hypertension was associated with the largest attenuation of the APO-CVD associations (except for GDM), followed by diabetes and hypercholesterolemia. For GDM, the association with CVD was substantially reduced after adjustment for postpregnancy diabetes; although the estimated proportion of attenuation was high (92.4%), the wide CI (0%–100%) indicates that the magnitude of mediation cannot be precisely quantified.

Contribution of APOs to CVD Risk Prediction

Characteristics of participants' established CVD risk factors are provided in Table S4. In Cox models predicting CVD from age 30 or 40 years, most established risk factors had statistically significant HRs regardless of APO

inclusion (Table S5). HRs for GHTN, preeclampsia, count of their episodes, and count of any APO episodes were slightly higher when predicting from age 30 years than from age 40 years.

As for predictive performance, all models demonstrated good calibration (Table 4). When predicting from age 30 years, adding APOs modestly improved discrimination beyond established CVD risk factors. Expanded Model 1 (adding GHTN and preeclampsia) slightly increased the C statistic (Δ C statistic: 0.005 [0.001–0.009]) but did not enhance risk reclassification overall (net reclassification index: –0.0002 [–0.011 to 0.012]) or among participants with or without CVD (Figure S6). Similar results were attained by Expanded Model 2 (adding GHTN, preeclampsia, and their count) and Expanded Model 3 (adding the count of any APO). For predictions from age 40, APOs had minimal impact on discrimination or risk reclassification overall and among participants with CVD. A small improvement in the C statistic was seen with Expanded Model 3 (Δ C statistic: 0.003

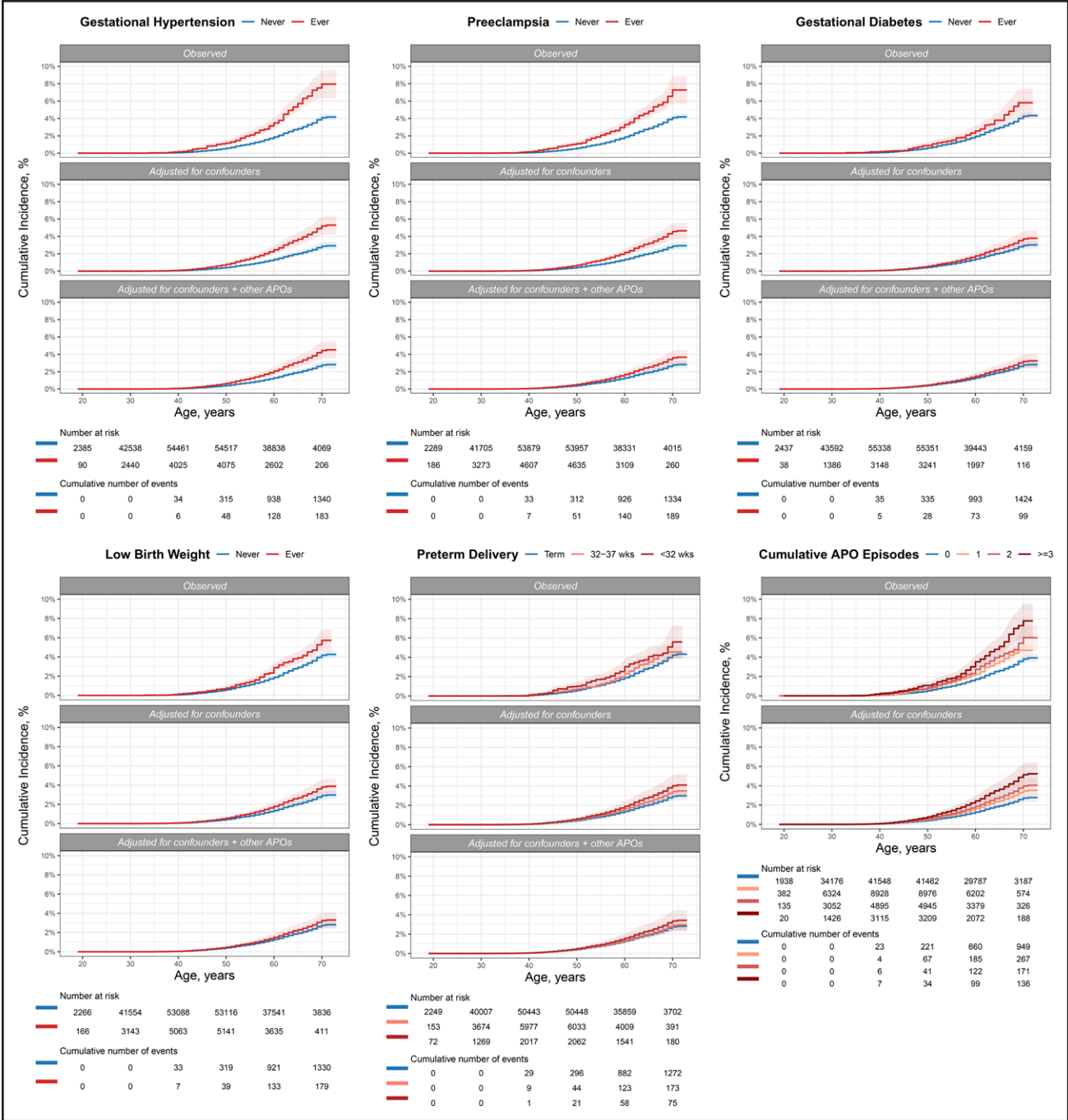


Figure. Observed and adjusted cumulative incidence of cardiovascular diseases over age, stratified by adverse pregnancy outcomes (APOs). Observed cumulative incidences were estimated from Kaplan-Meier survival functions. Cumulative incidences adjusted for confounders were estimated using Cox models in which continuous and categorical confounding variables were set to their mean and mode values, respectively. These included participants' mother's and father's education (12 years), parental history of myocardial infarction or stroke before age 60 (no), prepregnancy body mass index (21.8 kg/m²), prepregnancy smoking (never), prepregnancy alcohol intake (3.4g/d), prepregnancy high cholesterol (no), strenuous activity at ages 18 to 22 (1–3 months/y), and parity (1). Cumulative incidences adjusted for confounders and other APOs were estimated from Cox models that additionally included all APOs, with the values of APOs other than the one of interest set to never. See Figure S4 for cumulative incidences of gestational hypertension in pregnancies occurred before vs in or after 1985.

[0.0002–0.007]). Among those without CVD, Expanded Model 1 (net reclassification index: 0.002 [0.0002–0.004]) and Model 2 (0.003 [0.002–0.005]) slightly improved risk reclassification.

Sensitivity Analyses

Sensitivity analyses limited to the first pregnancy or accounting for immortal person-time yielded similar

Table 3. Attenuation of the Association Between Adverse Pregnancy Outcome in First Pregnancy and Incident Cardiovascular Disease by Adjustment for Intermediates Diagnosed After First Pregnancy, Including Hypertension, Diabetes, and Hypercholesterolemia

Adverse pregnancy outcomes	n (%)	Cases/ person-years	Hazard ratio (95% CI)		Proportion attenuated (95% CI)*			
			Without intermediates	With intermediates [†]	Jointly by 3 intermediates	By hypertension	By diabetes	By hypercholesterolemia
Gestational hypertension								
No	54 776 (95.0%)	1374/1968 491	Reference	Reference	Reference	Reference	Reference	Reference
Yes	2888 (5.0%)	123/97 699	1.70 (1.42–2.05)	1.16 (0.96–1.40)	71.9% (41.6%–90.2%)	76.0% (42.1%–93.3%)	10.9% (6.2%–18.5%)	13.7% (8.2%–21.9%)
Preeclampsia								
No	53 935 (93.5%)	1339/1 935 840	Reference	Reference	Reference	Reference	Reference	Reference
Yes	3729 (6.5%)	158/130 350	1.63 (1.38–1.92)	1.21 (1.02–1.43)	60.8% (38.3%–79.4%)	58.4% (37.7%–76.5%)	11.5% (6.9%–18.6%)	11.2% (6.5%–18.6%)
Gestational diabetes								
No	56 071 (97.2%)	1447/2 014 854	Reference	Reference	Reference	Reference	Reference	Reference
Yes	1593 (2.8%)	50/51 336	1.30 (0.98–1.73)	0.99 (0.74–1.31)	Not available	36.9% (8.8%–78.0%)	92.4% (0.0%–100.0%)	44.3% (9.9%–85.3%)
Low birth weight								
No	53 930 (93.5%)	1364/1933,431	Reference	Reference	Reference	Reference	Reference	Reference
Yes	3118 (5.4%)	105/110 179	1.30 (1.06–1.58)	1.20 (0.98–1.46)	29.9% (10.9%–59.9%)	33.8% (12.8%–64.1%)	Not significant	Not significant
Preterm delivery								
Term	52 756 (91.5%)	1348/1 893 998	Reference	Reference	Reference	Reference	Reference	Reference
Preterm: 32–37 wk	3740 (6.5%)	99/130 260	1.06 (0.86–1.30)	1.03 (0.84–1.26)	51.1% (0.1%–100.0%)	Not significant	Not significant	Not significant
Preterm: <32 wk	1120 (2.0%)	47/40616	1.59 (1.30–1.95)	1.44 (1.17–1.76)	21.8% (12.4%–35.5%)	19.6% (11.4%–31.7%)	9.6% (5.2%–17.0%)	4.9% (1.7%–13.1%)
Any APO								
No	40 879 (70.9%)	937/1473 141	Reference	Reference	Reference	Reference	Reference	Reference
Yes	16 785 (29.1%)	560/593 049	1.40 (1.26–1.55)	1.15 (1.03–1.28)	58.4% (38.7%–75.8%)	50.8% (34.6%–66.9%)	21.3% (14.3%–30.4%)	10.2% (6.3%–15.9%)
Cumulative APO episodes								
None	45 925 (79.6%)	1085/1 656 394	Reference	Reference	Reference	Reference	Reference	Reference
1 episode	8165 (14.2%)	279/287 246	1.41 (1.23–1.60)	1.19 (1.04–1.36)	49.2% (29.9%–68.7%)	45.1% (28.1%–63.2%)	16.7% (10.1%–26.3%)	7.5% (3.6%–14.7%)
2 episodes	2867 (5.0%)	105/99 768	1.52 (1.33–1.73)	1.17 (1.03–1.34)	61.7% (40.4%–79.3%)	57.0% (38.4%–73.9%)	15.3% (10.1%–22.6%)	13.2% (8.6%–19.7%)
≥3 episodes	707 (1.2%)	28/22 782	1.87 (1.64–2.13)	1.25 (1.10–1.43)	64.0% (49.1%–76.6%)	59.3% (46.1%–71.2%)	14.4% (10.9%–18.9%)	19.3% (15.0%–24.5%)

This mediation analysis was restricted to 57 664 women after excluding those who had hypercholesterolemia, chronic hypertension, or diabetes before first pregnancy. Cox models used years since first pregnancy as the time scale and origin. All models adjusted for age at pregnancy (continuous), participants' mother's and father's education (<9, 9–11, 12, 13–15, ≥16 y [reference]), parental history of myocardial infarction or stroke before age 60 years (no [reference], yes), prepregnancy body mass index (continuous), prepregnancy smoking (never [reference], former, current), prepregnancy alcohol intake (continuous), and strenuous activity at ages 18–22 years (never, 1–3 [reference], 4–6, 7–9, 10–12 mo/y). Missingness: low birth weight 1.1%, preterm delivery 0.1%. APO indicates adverse pregnancy outcome; and CVD, cardiovascular disease.

* $P < 0.05$ for all proportion attenuated statistics unless noted. Individual proportion attenuated statistics may not sum to the joint proportion attenuated statistic and may exceed 100% caused by shared pathways, as women may develop multiple intermediates between a delivery and an CVD event and the presence of one intermediate may affect another. Joint proportion attenuated not available for gestational diabetes because the estimated statistic was not between 0% and 100%.

†Hazard ratios obtained from a model including all 3 intermediates.

Table 4. Comparison of Premature CVD Risk Prediction Performance Between Models With and Without Adverse Pregnancy Outcomes

Statistics	Base Model	Expanded Model 1	Expanded Model 2	Expanded Model 3
	Established risk factors	Established risk factors	Established risk factors	Established risk factors
		Parity, GHTN, preeclampsia	Parity, GHTN, preeclampsia, cumulative episodes of GHTN or preeclampsia	Parity, cumulative episodes of any APO
Prediction at ≥30 y of age				
Calibration slope	1.04 (0.98–1.10)	1.04 (0.98–1.10)	1.05 (0.98–1.11)	1.04 (0.99–1.10)
C statistic	0.660 (0.642–0.676)	0.664 (0.647–0.680)	0.665 (0.650–0.680)	0.666 (0.651–0.682)
ΔC statistic	Reference	0.005 (0.001–0.009)	0.006 (0.002–0.010)	0.006 (0.002–0.012)
NRI overall	Reference	−0.0002 (−0.011 to 0.012)	0.006 (−0.005 to 0.021)	0.005 (−0.011 to 0.021)
NRI among participants with CVD	Reference	0.001 (−0.010 to 0.012)	0.005 (−0.006 to 0.019)	0.009 (−0.007 to 0.025)
NRI among participants without CVD	Reference	−0.001 (−0.003 to 0.001)	0.002 (−0.0001 to 0.003)	−0.004 (−0.006 to −0.001)
Prediction at ≥40 y of age				
Calibration slope	1.02 (0.96–1.09)	1.03 (0.96–1.09)	1.02 (0.96–1.09)	1.03 (0.97–1.09)
C statistic	0.672 (0.658–0.686)	0.674 (0.660–0.689)	0.674 (0.659–0.689)	0.675 (0.660–0.691)
ΔC statistic	Reference	0.002 (−0.0004 to 0.005)	0.002 (−0.0004 to 0.005)	0.003 (0.0002 to 0.007)
NRI overall	Reference	−0.004 (−0.017 to 0.007)	−0.006 (−0.019 to 0.006)	−0.009 (−0.022 to 0.003)
NRI among participants with CVD	Reference	−0.006 (−0.018 to 0.005)	−0.009 (−0.022 to 0.003)	−0.007 (−0.019 to 0.004)
NRI among participants without CVD	Reference	0.002 (0.0002–0.004)	0.003 (0.002–0.005)	−0.002 (−0.004 to 0.0004)

Established risk factors included smoking, diabetes history, systolic blood pressure, hypertensive medication use, total cholesterol, and high-density lipoprotein that assessed at or before age 30 or 40 years, as well as age (which was used as the time scale in the Cox model). NRI was estimated based on risk categories cut at event rate. APO indicates adverse pregnancy outcome; CVD, cardiovascular disease; GHTN, gestational hypertension; and NRI, net reclassification index.

results. Excluding coronary revascularization from the composite CVD outcome slightly increased most effect estimates in the association analyses (Table S6), modestly decreased effect estimates in the mediation analyses (Table S7), and yielded similar estimates in the prediction analyses (Table S8), without altering our overall conclusions. For example, associations between most APOs and CVD were strengthened, while the associations for GDM were slightly attenuated (Table S6).

DISCUSSION

Using over 35 years of follow-up in ≈60 000 parous participants in the NHS II, our study provides new insights into how multiple types of APOs over the reproductive years relate to long-term CVD risk and their potential value for CVD risk stratification. Each APO—GHTN, preeclampsia, GDM, LBW, and PTD—was individually associated with higher CVD risk, but only GHTN and preeclampsia remained independently associated with CVD after accounting for cooccurrence with other APOs. Much of the observed higher CVD risk associated with APOs in the first pregnancy was explained by the subsequent development of hypertension, diabetes, and hypercholesterolemia. No statistically significant interactions between APOs were observed. In addition, when predicting premature CVD risk from age 30 years, GHTN,

preeclampsia, and cumulative APO episodes modestly improved model discrimination beyond established CVD risk factors. When prediction began at age 40 years, the addition of APOs made small improvements in reclassification among those without CVD. Given these small gains in prediction performance, the clinical utility of a CVD risk prediction model that simply adds APOs to established CVD risk factors appears limited.

Our study contributes to existing research on multiple APOs and future CVD risk, with findings on independent and joint associations that differ from earlier reports. For example, a study from the WHI in the United States found HDP and LBW to be independently associated with CVD,⁵ and a national Swedish study reported independent associations of GDM, HDP, and PTD with ischemic heart disease² and CVD mortality,⁸ as well as additive interactions. These discrepancies may stem from several key methodological differences. The WHI study included participants who had survived until 2017, at which time they had a mean age of 70 years and responded to the survey about APOs, raising concerns about survival and recall bias. In contrast, these concerns are reduced in our study, as NHS II collected APO history at around ages 45 to 62 years. Unlike our time-to-event analysis, the WHI study used logistic regression, which cannot account for censoring, incorporate event timing, or estimate HRs that reflect instantaneous risk. It also adjusted for variables

measured well after pregnancy, potentially overadjusting for intermediates in the APO-CVD association. As for the Swedish study, it was conducted in a population with different geographic and lifestyle characteristics and focused on outcomes (ie, ischemic heart disease and CVD mortality) that differ from those assessed in our analysis (ie, MI, stroke, coronary revascularization). Further, while we tested for interaction between different APOs, most joint exposure groups were small and the overall event rate was low, limiting statistical power to detect moderate interaction effects, which may have contributed to the null interaction findings in our study.

The modest cohort effect we observed for the association between GHTN and later CVD, where stronger association was seen among women whose GHTN-complicated pregnancies occurred before 1985, is novel. This may reflect historical changes in diagnostic practices and BP monitoring during pregnancy, as well as improvements in prenatal care, postpartum follow-up, and CVD prevention strategies over time. For example, BP was less routinely measured throughout gestation in earlier years, and clinicians may have more broadly applied the term pregnancy-related high blood pressure for cases that would later be classified as preeclampsia. Consequently, GHTN reports from earlier decades may reflect more severe cases, contributing to stronger associations with CVD. Future studies should consider temporal trends when estimating long-term APO-CVD associations using historical cohort data. Regardless, our main conclusions held despite the cohort effects.

In our attenuation-based mediation analysis, the substantial reduction in the APO-CVD association after adjustment for postpregnancy hypertension, diabetes, and hypercholesterolemia is consistent with prior NHS II findings^{19,30} and studies in Finland and the United Kingdom.³¹ However, because our analyses quantified statistical attenuation rather than counterfactual natural direct and indirect effects, the results should not be interpreted as evidence of causal mediation or mechanistic pathway effects of APOs on CVD risk factors or events. Nevertheless, these findings support the importance of early identification and management of hypertension and diabetes in women with a history of APOs.

To date, few studies have evaluated whether incorporating multiple types of APOs improved long-term CVD risk prediction,^{7,15} whereas most focused on single types of APOs.^{10–14} In the UK Biobank, adding eclampsia and GDM to risk models did not improve prediction for women aged 45 to 69 years, with a follow-up of 9.9 to 11.3 years.¹⁵ Similarly, a retrospective cohort study in the United States found that including HDP and GDM did not improve 5-year risk prediction for women aged 40 to 79 years.⁷ Compared with these earlier studies, our study offers several advantages. First, our cohort had an APO prevalence more reflective of the general population of reproductive age than the UK Biobank study (<0.1% for

eclampsia, 0.3% for GDM). Second, we had a considerably longer (mean of 35.6 years) follow-up period. More importantly, we compared the performance of models predicting from both younger and older ages. Our finding that APOs improved discrimination only when prediction began at a younger age aligns with our prior work on PTD,¹³ and may reflect stronger associations between APOs and CVD when both APOs and established risk factors were assessed earlier in life, before the established risk factors became more common and potentially mediate the APO-CVD association.

These findings suggest that incorporating APO history into risk estimation of premature CVD events immediately after pregnancy, rather than waiting until women reach their 40s when tools like the Pooled Cohort Equation-Atherosclerotic Cardiovascular Disease calculator²⁵ are typically applied, may help close a ≈ 10 -year gap in preventive care. However, the observed predictive improvements were small, and since the majority of observed CVD events were premature, longer follow-up is needed for additional events to accrue at older ages for meaningful comparisons of the potential added value of APOs in predicting early- and later-onset CVD. Thus, simply adding APOs to existing CVD risk calculators for women may be insufficient and inefficient to guide pharmacotherapy initiation or modify current risk calculator thresholds.

Findings from our study suggest that APOs may play complex roles in long-term CVD risk, contributing to the ongoing debate over whether APOs primarily reflect underlying cardiometabolic susceptibility present before pregnancy (ie, the stress test hypothesis) or whether APOs actively increase CVD risk.^{32,33} In the former model, the physiological stress of pregnancy temporarily reveals subclinical cardiometabolic risk trajectories present before the pregnancy; both the APO and its later cardiometabolic correlates are caused by the same underlying risk, and the APO serves as an early, potentially useful, risk marker of a previously undetected cardiometabolic risk trajectory.³³ Supporting this are studies showing that women who develop APOs had higher cardiometabolic risk before pregnancy,^{34–36} and a Mendelian randomization study³⁷ showing that the genes predicting APOs in women are equally predictive of future CVD in their brothers, who cannot experience an APO. On the other hand, prior sister studies² report persistent associations after partially accounting for shared genetic and early-life environmental factors, and some (though not all) animal models^{38,39} recapitulating the preeclampsia phenotype suggest long-term vascular, cardiac, and neurovascular alterations.

Our observational analyses cannot distinguish between these competing hypotheses regarding whether APOs primarily reveal underlying cardiometabolic susceptibility or contribute directly to later CVD

risk. Although we observed a dose-response association between the number of APOs across the reproductive lifespan and subsequent CVD, as well as attenuation of these associations after adjustment for later cardiometabolic risk factors, such patterns are compatible with either mechanism. Recurrent APOs could reflect accumulating cardiometabolic risk before successive pregnancies in someone on a steep cardiometabolic trajectory. Because our mediation analyses were attenuation-based, the observed statistical attenuation of the association should not be interpreted as evidence that APOs can accelerate long-term cardiometabolic pathophysiology. Regardless, the importance of APO history in CVD prevention should not be understated. At minimum, APOs could serve as actionable signals to trigger earlier postpartum surveillance and CVD preventive strategies, such as BP and glucose monitoring, as well as lifestyle interventions.

The primary limitation of our study is the reliance on self-reported data for APOs, confounding variables, and most established risk factors. Although APOs were reported retrospectively, sometimes decades after pregnancies, misclassification is likely minimal given that all participants were registered nurses. Prior validation studies in NHS II have shown good agreement between self-reported and medical record data for preeclampsia¹⁹ and hypertension.²² Nonetheless, we acknowledge that APO misclassification may remain, particularly for earlier pregnancies, and that diagnostic criteria for APOs such as preeclampsia and GHTN have evolved over time, which potentially affected how these conditions were understood and reported. Though we used predicted total cholesterol and HDL, prior studies demonstrated strong correlations with measured values.^{10,13} Nevertheless, any measurement error is likely to be nondifferential and independent from the measurement of the outcome, which would typically bias results toward the null. In addition, the use of self-reported data limited the level of detail available, especially for birth weight and gestational age, which were reported in categories, thus precluding more granular analyses.

Further, although our analysis depends on survival to 2009 when reproductive history was assessed, 98.2% of the NHS II participants were alive then, minimizing survival bias. However, selection bias is possible, as the 76% of NHS II participants who completed the 2009 questionnaire differed modestly from nonrespondents in several baseline characteristics, including race, parental education, CVD history, and smoking status. Finally, although NHS II is one of the few prospective longitudinal cohorts of women in the United States with long-term follow-up that enables the investigation of lifetime APO history in relation to future CVD risk, the generalizability of our findings may be limited. The cohort consists predominantly of White, educated registered nurses with relatively healthy lifestyle profiles compared with contemporary younger women, and ≈50% of pregnancies

occurred before 1985. As such, our study population may not reflect the sociodemographic, health, and reproductive profiles of more recent US cohorts. These factors may influence baseline risk and potentially impact the observed findings. Caution is therefore warranted when generalizing these findings to more contemporary and diverse populations.

CONCLUSIONS

Common APOs among parous women in the United States, including GHTN, preeclampsia, GDM, LBW, and PTD, were each individually associated with higher long-term CVD risk. Much of the observed higher CVD risk associated with APOs in first pregnancy was explained by the hypertension, diabetes, and hypercholesterolemia developed after the pregnancy. Among the APOs, GHTN, preeclampsia, and the cumulative episodes of any APO provided additional value in improving prediction of premature CVD risk beyond established CVD risk factors used in existing risk estimation tools. However, these improvements in risk prediction were small and evident mostly when prediction began at a younger age. Taken together, our findings suggest that APOs may function as risk enhancers that support earlier cardiovascular surveillance and prevention, rather than as pharmaceutical decision-altering predictors within existing CVD risk calculators. Future research should focus on identifying most effective strategies to integrate APOs into clinical risk prediction tools.⁴⁰ This may involve developing new equations tailored for women with APOs or shifting the focus toward predicting intermediate outcomes such as hypertension or diabetes, particularly in younger women.

PERSPECTIVES

By examining multiple APO types across the reproductive lifespan in one of the largest, longest-running, and well-characterized cohorts of women, we provide robust evidence that common APOs were significantly associated with long-term CVD risk. HDP, including GHTN and preeclampsia, showed the strongest independent associations with later CVD, even after accounting for cooccurrence of other APOs. Statistical mediation analysis further revealed that a substantial proportion of this risk was attenuated after adjusting for the subsequent development of hypertension, diabetes, and hypercholesterolemia, pointing to tangible intervention targets. These findings support greater clinical and public awareness of the long-term cardiovascular implications of APOs.

Our results further emphasize the need to expand CVD risk assessment to include reproductive history. While incorporating HDP into existing CVD risk calculators yielded only modest predictive improvements,

particularly when major CVD predictors such as systolic blood pressure and diabetes were already included, APO history may help identify at-risk women earlier, enabling proactive postpartum BP monitoring, lifestyle counseling, and cardiovascular prevention. Future efforts should also consider developing tailored prediction tools for women with APOs, shifting the focus toward short-term risk of hypertension or diabetes to better guide postpartum care.

As diagnostic criteria and clinical practices continue to evolve, future research should refine APO-based CVD risk stratification and assess mediation pathways in diverse and contemporary populations, and determine whether risk-based postpartum interventions improve long-term cardiovascular outcomes. Thoughtful integration of reproductive history into CVD prevention, alongside targeted postpartum care, could meaningfully improve the cardiovascular health trajectory for millions of women with APOs.

ARTICLE INFORMATION

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Supplemental Material

Tables S1–S8
Figures S1–S6

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